💪 PHYSICS CH.5 — FORCE AND PRESSURE — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📑 **Topics in this chapter:**

1. Archimedes' Principle & Buoyancy

- Archimedes' Principle: Fluid mein fully/partially immersed body par upward buoyant force = weight of displaced fluid. *[Q836, Q848, Q853, Q911, Q938]*
- Formula: $F_b = \rho \times g \times V$; ρ = liquid density, V = volume, g = gravity. *[Q848, Q938]*
- Applications: Submarines (dive/surface), Ships, Hydrometers (liquid density measure), Lactometers (milk purity). *[Q848, Q853]*
- Archimedes ne is principle se king's crown ki gold purity determine ki thi. *[Q853]*
- When object floats: upthrust = weight of object (equilibrium). *[Q911]*
- Buoyant force acts at centre of mass of displaced fluid. *[Q836]*
- Buoyant force depends on density of fluid (ρ) and volume displaced (V). *[Q938]*

2. Gas Laws: Boyle's, Charles's, Gay-Lussac's

- Boyle's Law: Constant temperature par, gas ka volume inversely proportional hota hai pressure se (PV =

constant). *[Q837]*

- Charles's Law: Constant pressure par, ideal gas ka volume directly proportional hota hai absolute temperature se. *[Q837]*
- Gay-Lussac's Law: Constant volume par, gas ka pressure directly proportional hota hai absolute temperature se. *[Q837]*
- Graham's Law: Constant pressure par, gas diffusion ki rate inversely proportional hoti hai molar mass ke square root se. *[Q837]*

3. Newton's Laws of Motion

- First Law (Law of Inertia): Body at rest → at rest; body in uniform motion → uniform motion (jab tak external force na lage). *[Q838, Q889, Q892, Q909, Q940, Q958, Q965]*
  - Examples: Bus start hone par backward jerk, coin on card flicked falls in tumbler, athlete ka run before jump. *[Q889, Q940]*
- Second Law: Rate of change of momentum directly proportional to applied force (F = ma). *[Q839, Q840, Q888, Q895, Q924, Q961, Q964]*
  - Momentum (p) = mass × velocity; vector quantity; unit = kg m/s. *[Q895, Q898, Q901, Q910, Q946, Q950, Q951]*
  - Acceleration = rate of change of velocity. *[Q896]*
- Third Law: Every action has equal and opposite reaction. *[Q887, Q893, Q906, Q929, Q932, Q960]*
  - Examples: Rocket launching (expels gases backward → moves forward). *[Q887, Q906]*
  - Force never occurs singly in nature. *[Q893, Q960]*
  - Action-reaction forces always act on different bodies, never on the same body — jaise 70 kg person ne 50 kg person ko 50 N se push kiya toh 50 kg person bhi 50 N ka hi reaction force lagayega (magnitude equal, chahe masses alag hon). *[Q929, Q985]*
- Force (F) = ma — quantitative definition from Second Law. *[Q961, Q964]*

4. Friction: Types, Order, Applications & Reduction

- Friction = force resisting relative motion between two surfaces in contact; acts opposite to motion. *[Q842, Q846, Q850, Q882, Q883, Q894, Q962]*
- Friction ke bina mushkil ho jaane wale kaam: wet marble floor par chalna (grip minimal ho jaati hai) — normal walking, door movement, carrom, box move karna friction ki wajah se easy hote hain. *[Q849]*
- Order of Frictional Force: Static > Sliding (Kinetic/Dynamic) > Rolling. *[Q842, Q852, Q855, Q861, Q875]*
  - Static = strongest; Rolling = least. *[Q842]*
- Static Friction: Object at rest; due to interlocking of irregularities between surfaces. *[Q847, Q869]*
- Sliding (Kinetic/Dynamic) Friction: Surfaces in relative motion/sliding. *[Q842, Q875, Q886]*
- Rolling Friction: Object rolls on surface; smaller than sliding friction. *[Q842, Q855, Q864]*
- Fluid Friction (Drag): Exerted by fluids (air/water) on moving objects; depends on speed, nature of fluid, shape of object. *[Q842, Q862, Q868, Q878, Q879, Q880]*
- Friction cannot be totally eliminated — microscopic irregularities always remain. *[Q861]*
- Friction ka formula: f = μN (μ = coefficient of friction, N = normal force). *[Q851, Q962]*
- Friction wears out materials: screws, ball bearings, shoe soles. *[Q851]*
- Without friction → object can never stop (Newton's first law). *[Q852]*
- Surfaces ko rough banana (grooves, treads). *[Q854, Q860, Q863]*
- Harder pressing of surfaces. *[Q865]*
- Cricket bowler ball ko thigh par rubs karta hai (polishing) — isse ek side smooth aur doosri rough ho jaati hai,

differential friction/air resistance se ball swing karti hai — grip better hoti hai. *[Q881]*

- Examples: Tyre treads (better grip), brake pads, gymnasts' chalk, shoe soles with grooves. *[Q854, Q860, Q863, Q865, Q962]*
- Lubricants: Oil, grease, graphite — thin film banate hain between surfaces. *[Q856, Q857, Q858, Q866, Q877]*
- Ball bearings: Replace sliding friction with rolling friction. *[Q872]*
- Air cushion: Between moving parts. *[Q871]*
- Polishing, streamlining. *[Q860]*
- Examples: Luggage with rollers, ball bearings in ceiling fans/bicycles. *[Q864, Q872]*

5. Pressure: Definition, Units & Atmospheric Pressure

- Pressure (P) = Force/Area = F/A = Thrust per unit area; unit = Pascal (Pa) = N/m^2. *[Q873, Q885, Q918, Q921, Q952, Q955]*
- P = F/A → pressure inversely proportional to area. *[Q918, Q952]*
- Nail ki nok nukti → small area → larger pressure (P = F/A). *[Q873]*
- Snowshoes → large area → smaller pressure on snow. *[Q952]*
- Knife blades sharp (small area) → high pressure for cutting. *[Q952]*
- Thrust = force acting perpendicular to surface; T = v(dm/dt). *[Q870, Q925]*
- ATM = unit of measuring pressure; 1 ATM = 101325 Pa = 760 mm Hg = 14.7 psi. *[Q884]*
- Atmospheric pressure = force exerted by weight of Earth's atmosphere on surface. *[Q913]*
- Barometer = measures air pressure; sudden decrease → stormy weather indicated. *[Q914]*
- Stress = restoring force per unit area = F/A; unit = N/m^2 or Pascal. *[Q900]*
- Sound waves mein compressions un jagah bante hain jahan air pressure high hoti hai (rarefaction = low pressure/density regions). *[Q899]*

6. Scalar vs Vector Quantities

- Scalar: Only magnitude — distance, speed, mass, temperature, energy, work, volume, area, time. *[Q843]*
- Vector: Magnitude + direction — force, torque, displacement, velocity, momentum, acceleration. *[Q843, Q898, Q919]*

7. Fundamental Forces

- Gravitational force = weakest of four fundamental forces. *[Q844]*
- Nuclear force = strongest fundamental force. *[Q844]*
- Electric force = interaction between charged bodies. *[Q844]*
- Buoyant force = upward force by fluid on immersed object. *[Q844]*
- Moon Earth ke around near-circular orbit mein centripetal force ki wajah se move karta hai — ye gravitational attraction se milta hai jo Moon ko curved path mein rakhta hai (centrifugal, mechanical, ya frictional force nahi). *[Q859]*

8. Kepler's Laws of Planetary Motion

- Law of Orbits: All planets move in elliptical orbits, Sun at one focus. *[Q845]*
- Law of Areas: Line joining planet to Sun sweeps equal areas in equal intervals of time. *[Q845]*
- Law of Periods: Square of period ∝ cube of semimajor axis ($T^2 \propto r^3$). *[Q845]*

9. Spring Balance & Hooke's Law

- Spring balance = measures force/weight; works on Hooke's Law: $F = -kx$. *[Q841, Q867, Q874]*
- k = spring constant, x = displacement from rest position. *[Q841]*
- Coiled spring stretches when force applied → pointer moves on graduated scale. *[Q867, Q874]*
- Types of springs: helical, leaf, torsion, disc, conical volute, compression, extension, spiral. *[Q867, Q874]*

10. Work, Energy, Momentum & Equations of Motion

- Work (W) = Force × displacement × cosθ = F·d·cosθ. *[Q890, Q915, Q941]*
  - θ = 0° → positive work; θ = 90° → zero work; θ = 180° → negative work. *[Q890, Q916, Q941]*
- Energy: Capability to do work; forms = potential (P.E = mgh) + kinetic (K.E = $mv^2/2$); unit = Joule. *[Q902]*
- Momentum (p) = mass × velocity = mv; vector; unit = kg m/s. *[Q898, Q901, Q910, Q946, Q950, Q951]*
  - Conservation of Momentum: Total momentum of closed system remains constant. *[Q950]*
  - In collision → momentum is always conserved. *[Q950]*
- Equations of Motion (constant acceleration, straight line):
  - $v = u + at$ → Velocity-Time relation (First equation). *[Q897, Q923]*
  - $s = ut + (1/2)at^2$ → Position-Time relation (Second equation). *[Q903, Q908]*
  - $v^2 = u^2 + 2as$ → Position-Velocity relation (Third equation). *[Q908, Q934]*
- Velocity = rate of change of displacement with time; vector. *[Q896, Q919]*
- Speed = rate of change of distance with time; scalar. *[Q919, Q930]*
- Acceleration = rate of change of velocity with time. *[Q896, Q957]*
- Average speed/average velocity = explain motion during a given time interval. *[Q917, Q930]*
- Uniform acceleration: velocity increases by equal amounts in equal intervals of time. *[Q947, Q953]*
- Positive acceleration → velocity increases; Negative acceleration (retardation) → velocity decreases. *[Q891, Q905, Q937, Q947]*
- Zero acceleration → body has constant velocity (not necessarily zero). *[Q907, Q954]*
- Uniform speed par acceleration = zero. *[Q907]*
- Two equal forces opposite directions → net force = zero. *[Q904]*
- Velocity-time graph ka slope = acceleration; area under curve = displacement. *[Q933]*

11. Stress, Thrust & Tension

- Stress = restoring force per unit area = F/A; unit = N/m^2 or Pascal; depends on area. *[Q900]*
- Thrust = force acting perpendicular to surface; T = v(dm/dt); unit = Newton. *[Q870, Q925]*
- Tension = force transmitted when rope/string pulled from both ends; T = mg. *[Q927]*
- Perseverance/Rigidity = property to maintain shape under external force. *[Q912]*

12. Types of Motion

- Uniform motion: Equal distances in equal time intervals. *[Q935, Q945]*
  - Examples: Airplane flying at constant speed, train on straight tracks. *[Q935]*
- Non-uniform motion: Different speeds in equal time intervals. *[Q935, Q945]*
  - Examples: Bus on crowded road, bike in traffic, 100m race, bouncy ball. *[Q935, Q945]*
- Uniform circular motion: Constant speed in circular path; velocity changes continuously (direction). *[Q931]*
- Curvilinear motion: Motion along curved path (projectile, planets around Sun). *[Q942]*
- Oscillatory motion: Back-and-forth motion about mean position. *[Q931]*

13. Inertia, Mass & Weight

- Inertia: Tendency to resist change in state of rest/motion; directly proportional to mass. *[Q876, Q892, Q920, Q926, Q958]*
  - Greater mass → greater inertia → greater force needed to accelerate. *[Q958]*
- Mass = amount of matter in object; constant (does not change with location); scalar; unit = kg. *[Q920, Q936]*
- Weight = force with which Earth attracts object; W = mg; vector; unit = Newton (N). *[Q924, Q959, Q963]*
  - Weight varies with location (gravity changes); mass remains constant. *[Q959]*
- SI units: Mass = kg, Length = m, Time = s, Current = A, Temperature = K, Amount = mol, Luminous intensity = cd. *[Q963]*

14. Balanced vs Unbalanced Forces

- Balanced force: Net force = zero → no acceleration → no change in speed/direction. *[Q928, Q949]*
  - Can still cause change in shape (deformation/compression/stretching). *[Q949]*
- Unbalanced force: Required to change speed or direction of motion. *[Q949]*
- Object moves with constant speed jab force negligible ho (Newton's first law). *[Q928]*

15. Impulse & Safety Applications

- Impulse = Force × Time = change in momentum. *[Q943]*
- Helmet increases impact time → reduces force on head (for same momentum change). *[Q943]*
- Formula: Impulse = F × t; for same Δp, increasing t reduces F. *[Q943]*

16. Miscellaneous Concepts

- Liquefaction of gases: Decrease temperature + increase pressure. *[Q922]*
- Hot dry day: High temperature + low humidity → increased evaporation → cooling. *[Q922]*
- Displacement: Shortest distance between initial and final positions; measured along straight line; vector. *[Q956]*
  - Can be zero even if distance travelled is non-zero (return to start). *[Q956]*
- Ball thrown vertically upward: Initial velocity positive upwards (by convention). *[Q948]*
  - As rises → gravity decelerates → positive velocity reduces → zero at highest point → negative as falls. *[Q948]*

- Muscles stretching ka example: flag hoisting — jab flag ko upar khinchte hain, rope ke saath saath arms/shoulders/back ki muscles bhi elongate/stretch hoti hain (door open karna ya box push karna stretching nahi hai). *[Q944]*

17. ⚠ Traps / Formula Cheat Sheet (MCQ Q836–Q965)

- ⚠ Trap: Archimedes' Principle = buoyant force; Pascal's Law = pressure transmitted equally in fluid; Bernoulli's Principle = speed increase → pressure decrease. *[Q836]*
- ⚠ Trap: Boyle's Law = PV = constant (constant T); Charles's = $V \propto T$ (constant P); Gay-Lussac's = $P \propto T$ (constant V). *[Q837]*
- ⚠ Trap: Friction order: Static > Sliding > Rolling (static strongest, rolling least). *[Q842]*
- ⚠ Trap: Static friction > sliding friction (not equal, not less). *[Q875]*
- ⚠ Trap: Friction cannot be totally eliminated by polishing/lubrication. *[Q861]*
- ⚠ Trap: Rolling friction < sliding friction → luggage mein rollers use hote hain. *[Q864]*
- ⚠ Trap: Lubricants (oil/grease/graphite) reduce friction, increase nahi. *[Q857, Q866]*
- ⚠ Trap: Drag = fluid friction; air resistance = frictional force by air. *[Q878, Q879]*
- ⚠ Trap: Nail pointed → small area → more pressure ($P = F/A$). *[Q873, Q918]*
- ⚠ Trap: Newton's Third Law examples = rocket, equal reaction force chahe masses alag ho; First Law = inertia examples (bus jerk, coin in tumbler). *[Q887, Q889, Q940, Q985]*
- ⚠ Trap: Retardation = negative acceleration (not positive, not variable). *[Q905, Q937]*
- ⚠ Trap: Uniform speed → acceleration = zero. *[Q907]*
- ⚠ Trap: Work negative jab $\theta = 180°$ (force opposite to displacement). *[Q916, Q941]*
- ⚠ Trap: Momentum depends on both mass and velocity (not just one). *[Q910, Q946]*
- ⚠ Trap: Force never occurs singly in nature (action-reaction pair). *[Q893, Q960]*
- ⚠ Trap: Spring balance measures force, not mass directly. *[Q841]*
- ⚠ Trap: Barometer drop → stormy weather, not warm/cold/pleasant. *[Q914]*
- ⚠ Trap: Action-reaction forces act on different bodies, never same body. *[Q929]*
- ⚠ Trap: Zero acceleration → constant velocity (not necessarily zero velocity). *[Q954]*
- ⚠ Trap: Balanced force → can cause change in shape but not motion. *[Q949]*
- ⚠ Trap: Weight = vector (magnitude + direction); Mass = scalar. *[Q959, Q963]*
- ⚠ Trap: Weight unit = Newton; Mass unit = kg. *[Q963]*
- ⚠ Trap: Force definition = Second Law ($F = ma$), not First or Third. *[Q964]*
- ⚠ Trap: Velocity-time graph slope = acceleration; area = displacement. *[Q933]*
- ⚠ Trap: Instantaneous velocity = average velocity jab zero acceleration ho. *[Q939]*
- ⚠ Trap: Inertia $\propto$ mass (not velocity, not force). *[Q920, Q926, Q958]*
- ⚠ Trap: Positive acceleration → velocity increases; negative → decreases. *[Q947, Q937]*
- ⚠ Trap: Displacement measured along straight line (not curved path). *[Q956]*
- ⚠ Trap: Compression = high pressure/density region; rarefaction = low pressure/density region (sound wave). *[Q899]*

18. Numerical: Newton's Second Law (F = ma)

- Force ka formula $F = ma$ hota hai; acceleration $a = (v-u)/t$ se nikalta hai. *[Q966, Q969, Q976, Q984, Q986, Q987, Q989, Q994, Q995, Q1000, Q1004, Q1036, Q1045, Q1055, Q1056, Q1057]*

- Agar object constant velocity se chal raha hai toh acceleration zero aur force bhi zero hota hai. *[Q970]*
- Retarding force ke case mein force ka direction motion ke opposite hota hai, isliye angle 180° hota hai. *[Q974]*
- Example: 10 kg ki body ka velocity 2 sec mein 5 m/s se 10 m/s badhe toh $a=(10-5)/2=2.5\ m/s^2$ aur F=25 N; 5 sec aur lagane par final velocity 17.5 m/s hoti hai. *[Q966]*
- Rest se 2 sec mein 5 m/s speed pakadne wali body ka acceleration = $(5-0)/2 = 2.5\ m/s^2$. *[Q969]*
- 2 kg body 4 m/s^2 acceleration se chale toh net force $F = 2\times4 = 8$ N. *[Q976]*
- 8 kg body ka velocity 4 se 9 m/s 2 sec mein badhe toh F=20 N hota hai. *[Q984]*
- Bullet (12 g) barrel se 0.006 sec mein 600 m/s velocity se nikle toh force $F=(m\times v)/t = (0.012\times600)/0.006 = 1200$ N. *[Q986]*
- Dusra similar bullet example (12 g, 0.006 sec, 300 m/s): $a=(300-0)/0.006=50000\ m/s^2$, $F=ma=0.012\times50000=600$ N. *[Q987]*
- 100 kg body par 200 N force laga toh 6 sec mein velocity 5 se 17 m/s ho jaati hai. *[Q989]*
- 500 kg ki car par 350 N force laga toh acceleration 0.7 m/s^2 milta hai. *[Q971]*
- 20 kg block par 80 N force laga toh acceleration 4 m/s^2 hota hai. *[Q1036]*
- 1000 kg ki car ko 20 m/s se 4 sec mein rokne ke liye 5000 N ka retarding force chahiye. *[Q1045]*
- 3 kg mass par 21 N force laga toh acceleration 7 m/s^2 hota hai. *[Q1055]*
- 10 kg mass par 20 N force laga toh acceleration 2 m/s^2 hota hai. *[Q1056]*
- 10 kg mass par 10 N force laga toh acceleration 1 m/s^2 hota hai. *[Q1057]*
- 100 kg body par 200 N force laga toh acceleration = 200/100 = 2 ms^{-2} (unit hamesha ms^{-2}, ms^{-1} nahi). *[Q1004]*
- Agar same force F do alag masses ko alag acceleration a_1 aur a_2 deti ho, toh combined mass pe acceleration $(a_1\times a_2)/(a_1+a_2)$ hoti hai. *[Q1058]*

19. Numerical: Equations of Motion

- Pehli equation: $v = u + at$ — velocity aur time ka relation. *[Q966, Q979, Q982, Q993, Q994, Q1002, Q1006, Q1008, Q1020, Q1022, Q1025, Q1027, Q1030, Q1038, Q1039, Q1044, Q1048]*
- Dusri equation: $s = ut + ½at^2$ — displacement aur time ka relation. *[Q968, Q975, Q999, Q1005, Q1018, Q1021, Q1028, Q1034]*
- Teesri equation: $v^2 = u^2 + 2as$ — velocity aur displacement ka relation bina time ke. *[Q998, Q1024]*
- Free fall mein hamesha initial velocity u = 0 aur acceleration a = g = 10 m/s^2 (ya 9.8) lete hain. *[Q968, Q975, Q998, Q999, Q1006, Q1018, Q1020, Q1022, Q1025, Q1034, Q1044]*
- Upward throw mein acceleration negative (-g) hota hai kyunki gravity motion ko oppose karti hai. *[Q978, Q1002, Q1026, Q1039, Q1040]*
- Upar fekne par highest point par velocity momentarily zero hoti hai. *[Q1026]*
- 0.8 sec free fall mein body 3.2 m niche girti hai. *[Q975]*
- 4 sec free fall mein body 80 m niche girti hai. *[Q999]*
- 8 sec mein acceleration 3 m/s^2 se body 96 m travel karti hai. *[Q1034]*
- 20 m height se girne par body ki final velocity 20 m/s hoti hai. *[Q998]*
- 25 m/s se upar fekne par body ko top tak pahunchne mein 2.5 sec lagte hain. *[Q1002]*
- 0.5 sec free fall ke baad velocity 5 m/s hoti hai. *[Q1006]*
- 0.7 sec, 0.8 sec, 0.9 sec, aur 1 sec free fall ke baad velocities 7, 8, 9, aur 10 m/s hoti hain respectively. *[Q1020, Q1022, Q1025, Q1044]*
- 10 m/s se start karke 5 sec mein 20 m/s velocity badhne ke liye 2 m/s^2 acceleration chahiye. *[Q1027]*
- 10 sec mein 200 m travel karne ke liye rest se 4 m/s^2 acceleration chahiye. *[Q1028]*
- 1 m/s^2 acceleration se 120 sec mein velocity 120 m/s ho jaati hai. *[Q1030]*
- Ball upar fekne par total time agar 13.5 sec ho toh initial velocity 67.5 m/s hoti hai (t_{up} = 6.75 sec). *[Q1039]*

- Mount Everest par g ka value kam hota hai standard 9.8 se kyunki doori zyada hai. *[Q1040]*
- Rocket ka height equation $h = 119t - 7t^2$ ho toh wapas ground par 17 sec baad aayegi. *[Q1005]*

20. Numerical: Distance in nth Second

- nth second mein travel ki hui doori ka formula: $s_n = u + a(n - ½)$ ya $s_n = u + (a/2)(2n-1)$. *[Q968, Q1029]*
- Rest se 4th second mein (a=10) body 35 m travel karti hai. *[Q968]*
- Rest se 3rd second mein (a=4/3) body 10/3 m travel karti hai. *[Q1029]*

21. Numerical: Conservation of Momentum

- Closed system mein total momentum hamesha conserved rehta hai: initial momentum = final momentum. *[Q967, Q980, Q981, Q1009, Q1011, Q1013, Q1014, Q1017]*
- Gun initially at rest ho toh total initial momentum zero hota hai; bullet aage jaati hai aur gun peche khisakti hai. *[Q967, Q1011, Q1013, Q1014, Q1017]*
- Recoil velocity ka formula: v_gun = -(m_bullet × v_bullet) / m_gun; negative sign opposite direction batata hai. *[Q967, Q1011, Q1013, Q1014, Q1017]*
- 0.04 kg shell 90 m/s se fire ho 120 kg gun se toh recoil speed 3×10^{-2} m/s hoti hai. *[Q967]*
- 0.02 kg bullet 150 m/s se 3 kg pistol se toh recoil 1.0 m/s hoti hai. *[Q1011]*
- 0.025 kg bullet 150 m/s se 2 kg pistol se toh recoil ≈ 1.87 m/s hoti hai. *[Q1013]*
- 0.03 kg bullet 150 m/s se 2 kg pistol se toh recoil 2.25 m/s hoti hai. *[Q1014]*
- 0.03 kg bullet 150 m/s se 3 kg pistol se toh recoil 1.5 m/s hoti hai. *[Q1017]*
- ⚠️ Trap: Q1009 mein official answer key mein values mismatch thi; exam mein official key follow karo. *[Q1009]*
- Collision mein: $m_1u_1 + m_2u_2 = (m_1+m_2)v$ jab dono stick ho jaayein; jaise 2000 kg truck 10 m/s se 500 kg car se takra ke 8 m/s chale toh car ka mass 500 kg nikalta hai. *[Q980]*

22. Numerical: Weight (W = mg) & Moon Weight

- Weight ka formula W = mg hota hai; mass ka formula m = W/g. *[Q972, Q988, Q990, Q1007, Q1010, Q1012, Q1015, Q1016, Q1019, Q1023, Q1031, Q1032, Q1042, Q1049, Q1050, Q1051, Q1052]*
- Earth par g ≈ 9.8 m/s² (ya 10 m/s² agar question mein specify ho). *[Q972, Q988, Q990, Q1010, Q1049, Q1051]*
- 450 N weight wale body ka mass g=9.8 par ≈ 45.9 kg hota hai. *[Q972]*
- 20 kg body ka weight g=9.8 par 196 N hota hai. *[Q990]*
- 20 kg body ka weight g=10 par 200 N hota hai. *[Q1007]*
- 10 kg body ka weight g=10 par 100 N hota hai. *[Q1023]*
- 200 N weight wale body ka mass g=10 par 20 kg hota hai. *[Q1031]*
- 100 kg body ka weight g=9.8 par 980 N hota hai. *[Q1049]*
- 98 N weight wale body ka mass g=9.8 par 10 kg hota hai. *[Q1051]*
- Moon par weight = Earth weight / 6 kyunki g_moon = g_earth/6 hota hai. *[Q1001, Q1012, Q1016, Q1019, Q1032, Q1042, Q1050, Q1052]*
- Earth par 60 N weight wale body ka Moon par weight 10 N hota hai. *[Q1001, Q1012, Q1019, Q1032]*
- Earth par 20 N weight wale body ka Moon par weight ≈ 3.33 N hota hai. *[Q1042]*
- Moon par 300 N weight wale body ka Earth par weight 1800 N hota hai. *[Q1052]*
- Mass Moon par bhi same rehti hai — sirf weight badalta hai. *[Q1033]*

- Earth weight hamesha Moon weight ka 6 guna hota hai. *[Q1016, Q1050]*

23. Numerical: Momentum (p = mv)

- Momentum ka formula $p = mv$ hota hai; unit kg m/s. *[Q977, Q983, Q991, Q992, Q996, Q1037, Q1041, Q1043]*
- 80 kg rider 60 m/s se ja raha ho toh momentum 4800 kg m/s hota hai. *[Q977]*
- Mass aadhi aur velocity double karne par momentum same (mv) rehta hai. *[Q983]*
- 200 g (0.2 kg) ball jiski kinetic energy 10 J ho uska momentum 2 kg m/s hota hai (pehle v=10 m/s nikalo). *[Q991]*
- 10 kg body 2 m/s se ja rahi ho toh momentum 20 kg m/s hota hai. *[Q992]*
- Momentum 50 aur velocity 5 ho toh mass 10 kg nikalti hai. *[Q1037]*
- 50 kg body 6 m/s se ja rahi ho toh momentum 300 kg m/s hota hai. *[Q1041]*
- 50 kg body 20 m/s se ja rahi ho toh momentum 1000 kg m/s hota hai. *[Q1043]*
- Same kinetic energy wali bodies mein momentum $\sqrt{m}$ ke proportional hota hai; mass ratio 1:9 ho toh momentum ratio 1:3 hota hai. *[Q1046]*

24. Numerical: Unit Conversions

- km/h ko m/s mein convert karne ke liye 5/18 se multiply karte hain. *[Q973, Q993, Q1008, Q1027, Q1048]*
- 90 km/h = 25 m/s; 72 km/h = 20 m/s; 54 km/h = 15 m/s; 36 km/h = 10 m/s; 18 km/h = 5 m/s. *[Q973, Q993, Q1008, Q1027, Q1048]*

25. Numerical: Work Done & Energy

- Work ka formula $W = Fd\cos\theta$ hota hai. *[Q974, Q1054]*
- Agar 20 N force se 2 m displacement mein 20 J work ho toh force aur displacement ke beech angle 60° hota hai ($\cos\theta = 1/2$). *[Q1054]*
- Spring compression mein block ki kinetic energy spring ki potential energy mein convert hoti hai: $\frac{1}{2}mv^2 = \frac{1}{2}kx^2$. *[Q1047]*
- 2 kg block 4 m/s se chal ke spring ko compress kare toh k=10000 N/m ke liye compression ≈ 5.5 cm hoti hai. *[Q1047]*

26. Numerical: Gravitational Force

- Do masses ke beech gravitational force $F = Gm_1m_2/r^2$ se milti hai; $G = 6.673\times10^{-11}$ Nm^2/kg^2. *[Q1053]*
- 20 kg aur 50 kg ke beech 2 m doori par force 166.75×10^{-10} N hoti hai. *[Q1053]*

27. Numerical: Cricket Ball / Retarding Force

- Cricket ball ko rokne par lagne wala retarding force $F = m(v-u)/t$ se nikalta hai. *[Q1035]*
- 100 g (0.1 kg) ball 30 m/s se 0.05 sec mein ruke toh -60 N ka retarding force lagta hai. *[Q1035]*

28. ⚠️ Traps / Formula Cheat Sheet (Numerical Q966–Q1058)

- ⚠️ Trap: Acceleration ka unit hamesha m/s^2 hota hai; m/s (velocity) nahi. *[Q971, Q982, Q1034]*
- ⚠️ Trap: Force ka unit Newton (N) hota hai; Pa, Joule, kg, ya m/s^2 nahi. *[Q989, Q997, Q1003, Q1031]*
- ⚠️ Trap: Momentum ka unit kg m/s hota hai; kg m/s^2 (jo N hota hai) nahi. *[Q977, Q992, Q1041]*
- ⚠️ Trap: Weight ka unit Newton hota hai; kg mass ka unit hai. *[Q1007, Q1023, Q1031, Q1049]*
- ⚠️ Trap: Moon par weight 1/6 ho jaata hai lekin mass same rehti hai. *[Q1016, Q1033, Q1050, Q1052]*
- ⚠️ Trap: Recoil velocity hamesha negative hoti hai kyunki direction opposite hoti hai. *[Q967, Q1011, Q1013, Q1014, Q1017]*
- ⚠️ Trap: nth second ki doori ke liye formula $s_n = u + a(n - ½)$ use hota hai; direct multiply mat karna. *[Q968, Q1029]*
- ⚠️ Trap: Upward throw mein acceleration -g lete hain (negative). *[Q978, Q1002, Q1039]*
- ⚠️ Trap: Highest point par velocity zero hoti hai, -g nahi. *[Q1026]*
- ⚠️ Trap: km/h → m/s conversion ke liye 5/18 se multiply karte hain, 18/5 se nahi. *[Q973, Q993, Q1048]*
- ⚠️ Trap: Q1009 mein official answer key mein values ka mismatch tha; solve karke official key se match karo. *[Q1009]*
- ⚠️ Trap: Work negative tab hota hai jab force aur displacement ke beech angle 90° se zyada ho. *[Q1054]*
- ⚠️ Trap: Gravitational constant $G = 6.673\times10^{-11}$; power 10^{-11} mat bhoolna. *[Q1053]*
- ⚠️ Trap: Spring compression ke numerical mein answer cm ya m mein dekh ke unit check karo. *[Q1047]*
- ⚠️ Trap: Ball upar fekne par total time = 2 × time to reach top. *[Q1039, Q1040]*